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6th Sem / Civil Engineering/Brick Technology/ Construction Management
Sub : Steel structures design and drawing

Time : 6Hrs.   M.M. : 150

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The IS code which deals with steel structure, is ______

a) IS : 456 b) IS : 800
c) IS : 1893 d) IS : 2000

Q.2 A channel section consist of _____

a) Two flanges
b) One web and two flanges
c) Two webs and two flanges
d) Two webs

Q.3 The most commonly used rivet head is _____

a) Snap head b) Pan head
c) Flat head d) All the above

Q.4 Two structural members may be temporarily connected by which kind of connection.

a) Welded b) Riveted
c) Bolted d) Pin

Q.5 Roof trusses are economical for spas________.

a) Greater than 3m b) Greater than 4m
c) Greater than 5m d) Greater than 6m

Q.6 Welded structures are ________

a) Lighter than riveted structure
b) Heavier than riveted structure
c) Similar to riveted structure
d) None of these

Q.7 Slenderness ratio has the unit as ______

a) mm b) mm^2

c) mm^3 d) Unit less

Q.8 In columns with both ends fixed, the effective length is _____

a) 1.2 L b) 1.5 L

c) 2.0L d) 0.65L

Q.9 Which of the following is not a type of column base _____

a) Slab base b) Gusseted base

c) Rolled base d) All the above

Q.10 The maximum spacing of vertical stiffeners in plate girder is limited to _______

a) d b) 2d

c) 0.8d d) 1.5d

SECTION-B

Note: Objective type questions. (select the appropriate option). All questions are compulsory. (10x1=10)

Q.11 The continuous deformation of a material under a constant load at high temperature is known as ______ (Creep / Brittleness)

Q.12 The maximum deflection, for a simply supported beam should not exceed ____. ((1/225 of span) / (1/325 of span)).

Q.13 The permissible longitudinal pitch in a riveted joint in tension is _____ ((12t or 200 mm whichever is less) / (16t or 200 mm whichever is less)).

Q.14 _____ = Load in the member / Rivet value (Number of rivets required / efficiency of Joint)

Q.15 Young's modulus of elasticity of steel is _____ ((2.0×10^5 N/mm^2) / (2.0×10^5 KN/mm^2)).

Q.16 The parallel fillet weld resists ____ (More strength / Less strength)

Q.17 If shorter legs are connected to the gusset plate, then strength will _____ (Increase / Decrease)

Q.18 When one end of column is hinged and other end is fixed, then effective length = _____ (($L/2$) / ($L/\sqrt{2}$)).

Q.19 The effective length of a column depends upon ____ (End conditions of the column / Load on a column)

Q.20 Expand the term ISLC.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Find the efficiency of a single riveted lap joint with the given data: Diameter of rivet = 22mm, pitch of rivets = 70mm, Thickness of plates = 12 mm, σ_{at} = 150 N/mm^2, τ_{vf} = 90 N/mm^2 and σ_{pf} = 270 N/mm^2.

Q.22 Calculate the safe load for a 8mm fillet welded joint with effective length of 150mm. The permissible shear stress in the weld is 108 N/mm^2.

Q.23 Enlist any five advantages of bolted connections.

Q.24 Write the different types of sections used as tension members along with sketches.

Q.25 Calculate the strength of single angle ISA 100mm x 75 mm x 8 mm used as a tie member with longer leg connected at ends by 14 mm dia. rivets on the gusset plate. Take F_{st} = 150 N/mm^2.

Q.26 Calculate the load carring capacity of ISMB 350 @ 514 N/m to be used as a column. The effective of the column is 3.5 m.

Q.27 Write the steps to be followed during the design of an axially loaded column.

Q.28 Name the main parts of a roof truss along with a sketch.

Q.29 How will you find out the economic spacing of roof trusses?

Q.30 Write a short note on splicing of columns.

Q.31 Describe the various causes of buckling in columns.

Q.32 A simply supported steel beam carries a superimposed load of 40.5kN/m over an effective span of 8.15 m. Design the beam and check for shear. Take permissible bending stress 165 N/mm^2, shear stress 100 N/mm^2 and E=2.1 x 10^5 N/mm^2.

Q.33 Write the assumptions made in the theory of simple bending.

Q.34 Draw the variation of shear stress with depth of a cross-section of abeam.

Q.35 Write a short note on "Erection of Girders" used as steel structures.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Determine the strength of an ISHT 75 which is used as a tie member. It is connected through its flange by means of 18mm diameter rivets. Assume σ_{at} = 150 Mpa.

Q.37 Calculate the safe axial load carried by a built-up coloum consisting of ISHB 400 @ 759.3 N/m with a plate 400x16mm is welded to each flange. The column is 4.0 m long and its effectively held in position at both ends but not restrained against rotation. Take f_y = 250 N/mm^2.

Q.38 a) Classify the column of the basis of mode of failure.

b) Write the procedure for design of tension members

SECTION-E

Note: Steel structure Drawing. Attempt any two questions out of three questions. (2x25=50)

Q.39 Draw the heel joint of a steel roof truss with roof drainage arrangement.

Q.40 Draw to a suitable scale the section plan, front elevation and cross section of a simple plate girder from the following given data.

- Clear span = 8m
- Web plate = 800 mm 12 mm
- Depth over angle = 810 mm
- Flange angle = 2-ISA 80x80x8 mm
- Top and Bottom cover plates = 200 mm x 12mm
- Bearing plate = 200 mm x 250 mm x 15 mm
- Diameter of rivets = 20 mm
- Pitch of rivets = 60 mm

Q.41 Draw the front and side elevations of seated beam to beam connection from the following data

- Main beam = ISLB 500 @ 735.8 N/m
- Secondary beam = ISLB 250 @ 273.7 N/m
- Seat angle = ISA 75x50x8mm
- Top cleat angle = ISA 65x45x8mm
- Diameter of rivets = 16mm